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Thermodynamic assessments of the Bi–Lu and Lu–Sb systems

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ABSTRACT

The phase diagrams and thermodynamic properties in the Bi–Lu and Lu–Sb binary systems have been assessed by using the CALPHAD (Calculation of Phase Diagrams) method on the basis of experimental data including the thermodynamic properties and phase equilibria. The Gibbs free energies of the liquid, hexagonal close-packed (hcp) and rhombohedral phases were described by the substitutional solution model, and the intermetallic compounds (Bi_2Lu, BiLu, Bi_3Lu_5, Lu_3Sb, Lu_5Sb_3, αLuSb, βLuSb and $LuSb_2$ phases) were treated as stoichiometric compounds. The thermodynamic parameters of the Bi–Lu and Lu–Sb binary systems were obtained, and agreement between the calculated results and experimental data was obtained for each binary system.

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1. Introduction

Magnesium (Mg) alloys have been widely used in many industrial fields due to their superior characteristics such as low density, high specific strength, and good castability [1]. The addition of rare earth (RE) elements to Mg-based alloys can enhance the high-temperature properties, corrosion resistance, and casting characteristics [2,3]. Besides, additions of Bi or Sb can improve the high-temperature properties of Mg-based alloys [4,5]. Therefore, it is important to understand the phase equilibria in the Mg–Bi–Sb–RE alloy system. In the present paper, as a part of the thermodynamic database of the Mg–Bi–Sb–RE alloy system, the thermodynamic descriptions for the phase diagrams in the Bi–Lu and Lu–Sb binary systems were obtained by means of the CALPHAD method, in which the thermodynamic parameters to describe the Gibbs free energy of the individual phases are optimized based on the experimental data of the thermodynamic properties and the phase diagram.

2. Experimental information

2.1. The Bi–Lu system

The Bi–Lu system consists of two terminal solution phases (rhombohedral (Bi) and hcp (Lu) phases) and three intermetallic compounds (Bi_2Lu, BiLu, and Bi_3Lu_5 phases). Based on differential thermal analysis, X-ray diffractometry, and metallography, Abulkhaev and Ganiev [6] investigated the phase diagram in the Bi–Lu system, and reported three stoichiometric compounds: Bi_2Lu, Bi_3Lu_5 (peritectic formation at 1450 ± 10 °C), and BiLu (melting point, 1820 ± 15 °C). Abulkhaev and Ganiev [6] also found two eutectic reactions: L (∼80 at.% Lu) ↔ Bi_3Lu_5 + (Lu) at 1165 ± 10 °C, and L (∼0.3 at.% Lu) ↔ (Bi) + BiLu at 269 ± 2 °C. In the work of Abulkhaev and Ganiev [6], the solid solubility of Bi in the hcp (Lu) phase was determined to be less than 1 at.%. The phase diagram of the Bi–Lu system reviewed by Okamoto [7] is shown in Fig. 1.

In addition, Colinet et al. [8] predicted the standard enthalpies of formation of the Bi_2Lu, BiLu, and Bi_3Lu_5 phases by using Miedema's model. With the same method, Colinet et al. [8] and Boer and Perrifor [9] predicted that the value of the enthalpy of mixing of the liquid phase with the reference states of pure liquid metals at 50 at.% Lu in the Bi–Lu system is respectively −42 kJ/mol [8] and −51 kJ/mol [9]; however, no temperature dependence was given in the work of Colinet et al. [8] and Boer and Perrifor [9].

2.2. The Lu–Sb system

The Lu–Sb system consists of two terminal solution phases (hcp (Lu) and rhombohedral (Sb) phases) and five intermetallic compounds (Lu_3Sb, Lu_5Sb_3, αLuSb, βLuSb, and $LuSb_2$ phases). Based on thermal analysis and microstructural analysis, Abdusalyamova et al. [10] investigated the phase diagram in the Lu–Sb system, and

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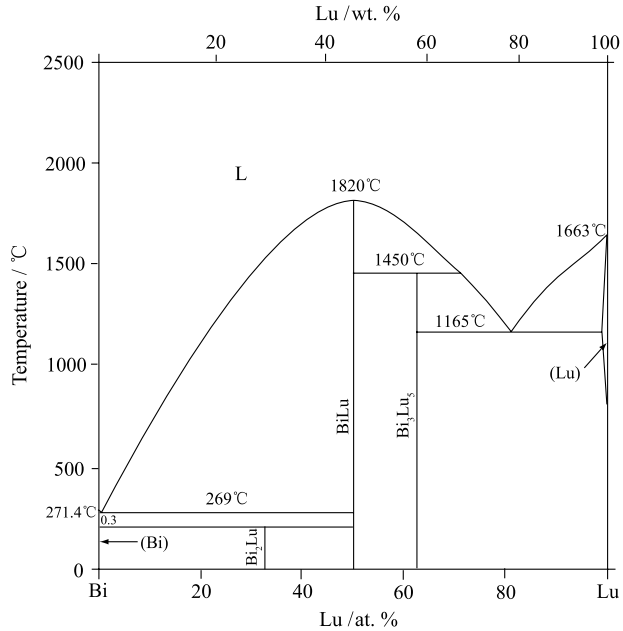

Fig. 1. The phase diagram of the Bi–Lu system reviewed by Okamoto [7].

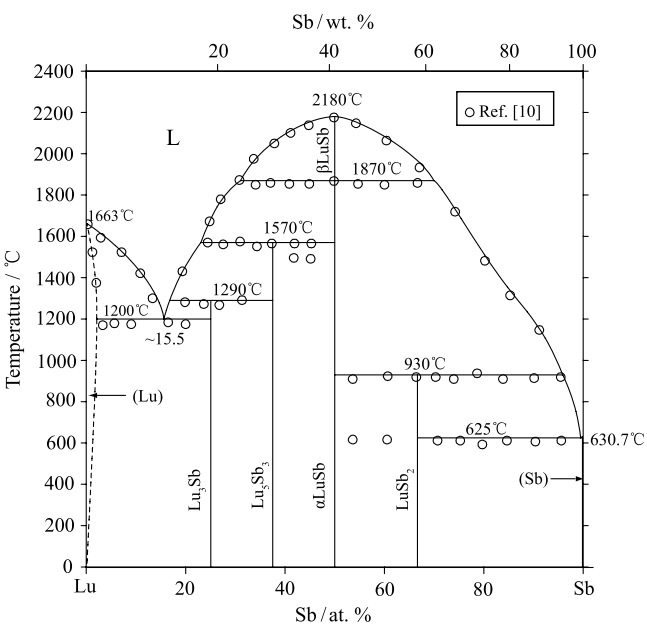

Fig. 2. The phase diagram of the Lu–Sb system reviewed by Okamoto [11].

reported four stoichiometric compounds: Lu_3Sb (peritectic formation at 1290 °C), Lu_5Sb_3 (peritectic formation at 1570 °C), LuSb (melting point, 2180 °C), and $LuSb_2$ (peritectic formation at 930 °C). Abdusalyamova et al. [10] also determined two eutectic reactions: L (∼15.5 at.% Sb) ↔(Lu) + Lu_3Sb at 1200 °C, and L (∼99 at.% Sb) ↔$LuSb_2$ + (Sb) at 625 °C. In Abdusalyamova's work [10], the solid solubility of Sb in the hcp (Lu) phase was suggested to be less than 2 at.%. A reversible polymorphic transition (αLuSb ↔ βLuSb at 1870 °C) was observed by Abdusalyamova et al. [10]. The phase diagram of the Lu–Sb system was reviewed by Okamoto [11] and is shown in Fig. 2.

In addition, using a simple direct-reaction calorimeter, Chua and Pratt [12] reported the standard enthalpy of formation of the αLuSb phase. Gerasimov et al. [13] determined the standard enthalpies and entropies of formation of the αLuSb and $LuSb_2$ phases. The measurement is actually the electromotive force (EMF). Colinet et al. [8] predicted the standard enthalpies of formation of the Lu_3Sb, Lu_5Sb_3, αLuSb and $LuSb_2$ phases by using Miedema's model. With the same method, Colinet et al. [8] and Boer and Perrifor [9] predicted that the value of the enthalpy of mixing of the liquid phase with the reference states of pure liquid metals at 50 at.% Sb in the Lu–Sb system is respectively −52 kJ/mol [8] and −66 kJ/mol [9]; however, no temperature dependence was given in the work of Colinet et al. [8] and Boer and Perrifor [9].

Table 1
The stable solid phases and the models used in the Bi–Lu [7] and Lu–Sb [11] systems.

System	Phase	Prototype	Modeling phase	Model used
Bi–Lu	(Bi)	αAs	$(Bi)_1$	SSM
	Bi_2Lu	–	$(Bi)_2(Lu)$	SM
	BiLu	NaCl	(Bi)(Lu)	SM
	Bi_3Lu_5	–	$(Bi)_3(Lu)_5$	SM
	(Lu)	Mg	$(Bi, Lu)_1 : (Va)_{0.5}$	SSM
Lu–Sb	(Lu)	Mg	$(Lu, Sb)_1 : (Va)_{0.5}$	SSM
	Lu_3Sb	–	$(Lu)_3(Sb)$	SM
	Lu_5Sb_3	Mn_5Si_3	$(Lu)_5(Sb)_3$	SM
	αLuSb	NaCl	(Lu)(Sb)	SM
	βLuSb	–	(Lu)(Sb)	SM
	$LuSb_2$	–	$(Lu)(Sb)_2$	SM
	(Sb)	αAs	$(Sb)_1$	SSM

SSM: Subregular solution model
SM: Sublattice model.

3. Thermodynamic models

Information about stable solid phases and the models used in the Bi–Lu and Lu–Sb binary systems is listed in Table 1.

3.1. Solution phases

In the Bi–Lu and Lu–Sb binary systems (which are both binary A–B systems), the Gibbs free energies of the liquid, hcp, and rhombohedral phases are described by the substitutional solution model, as follows:

$$G_m^{\phi} = {}^0G_A^{\phi}x_A + {}^0G_B^{\phi}x_B + RT(x_A \ln x_A + x_B \ln x_B) + {}^EG_m^{\phi}, \quad (1)$$

where ${}^0G_i^{\phi}$ is the Gibbs free energy of the pure component i in the respective reference state with ϕ phase, which is taken from the SGTE pure element database [14]; x_i denotes the mole fraction of component i; R is the gas constant; T is the absolute temperature; and the term ${}^EG_m^{\phi}$ represents the excess Gibbs free energy, which is expressed in Redlich–Kister polynomial form [15] as

$${}^EG_m^{\phi} = x_A x_B \sum_{m=0}^{n} {}^mL_{A,B}^{\phi}(x_A - x_B)^m, \quad (2)$$

$${}^mL_{A,B}^{\phi} = a + bT, \quad (3)$$

where ${}^mL_{A,B}^{\phi}$ is the binary interaction parameter, and the coefficients of a and b are evaluated on the basis of available experimental data.

3.2. Stoichiometric intermetallic compounds

All the intermetallic compounds in the Bi–Lu and Lu–Sb binary systems are treated as stoichiometric phases. The Gibbs free energy per mole of formula unit A_pB_q can be expressed as follows:

$$\Delta^0G_f^{A_pB_q} = G_{A:B}^{A_pB_q} - p^0G_A^{SER} - q^0G_B^{SER} = a' + b'T, \quad (4)$$

where $\Delta^0G_f^{A_pB_q}$ indicates the standard Gibbs free energy of formation of the stoichiometric compound from pure elements; the parameters a' and b' are evaluated in the present work.

Table 2
Experimental and calculated invariant reactions in the Bi–Lu system.

Invariant reaction	Reaction type	Composition (at.% Lu)			Temperature (°C)	Reference
L ↔ (Bi) + BiLu	Eutectic	0.3	0	50	269 ± 2	[6]
		0.8	0	50	269.6	This work
(Bi) + BiLu ↔ Bi_2Lu	Peritectoid	100	50.0	33.3	210	[7]
		100	50.0	33.3	210	This work
L ↔ BiLu	Congruent		50.0		1820 ± 15	[6]
			50.0		1819.8	This work
L + BiLu ↔ Bi_3Lu_5	Peritectic		50.0	62.5	1450 ± 10	[6]
		69.5	50.0	62.5	1450	This work
L ↔ Bi_3Lu_5 + (Lu)	Eutectic	80 ± 1	62.5	≥99	1165 ± 10	[6]
		82.4	62.5	99	1165.1	This work

Table 3
Thermodynamic parameters in the Bi–Lu system assessed in the present work.

Parameters in each phase (J/mol)
Liquid phase, format (Bi, Lu)
${}^0L^{Liq}_{Bi,Lu} = -184333 + 27.6T$
${}^1L^{Liq}_{Bi,Lu} = 8092$
${}^2L^{Liq}_{Bi,Lu} = 33980$
Hcp phase, format (Bi, Lu)
${}^1L^{Hcp}_{Bi,Lu} = 82216$
Bi_2Lu phase, format $(Bi)_{0.667}(Lu)_{0.333}$
$\Delta^0G_f^{Bi_2Lu} = -56000 + 19.555T$
BiLu phase, format $(Bi)_{0.5}(Lu)_{0.5}$
$\Delta^0G_f^{BiLu} = -72696 + 5.794T$
Bi_3Lu_5 phase, format $(Bi)_{0.375}(Lu)_{0.625}$
$\Delta^0G_f^{Bi_3Lu_5} = -63731 + 7.758T$

Table 4
Comparison of predicted and calculated values for the standard enthalpies of formation (kJ/mol of atoms) of intermetallic compounds in the Bi–Lu system.

Phase	Predicted [8]	Calculated
Bi_2Lu	−56	−56
BiLu	−72	−72.696
Bi_3Lu_5	−67	−63.731

The reference states of pure Bi and Lu are rhombohedral and hcp phases, respectively.

4. Optimized results and discussion

The optimization of the thermodynamic parameters was carried out by using the PARROT program in the Thermo-Calc software [16], which can handle various kinds of experimental data. The experimental data of the phase diagram and thermodynamic properties were used as input to the program. Each piece of selected information was given a certain weight based on the importance of the data, which was changed by trial and error during the assessment, until most of the selected experimental information was reproduced within the experimental uncertainty limits.

4.1. The Bi–Lu system

The calculated Bi–Lu phase diagram compared with the experimental data is shown in Fig. 3, and the calculated compositions and temperatures for the invariant reactions compared with the selected experimental data are listed in Table 2. It is seen that the calculated results are in reasonable agreement with the experimental data [6]. A set of complete self-consistent thermodynamic parameters describing the Gibbs free energy of each phase in the Bi–Lu system is given in Table 3.

The calculated standard enthalpies of formation of intermetallic compounds with reference states of the Bi (rhombohedral) and Lu (hcp) phases compared with the predicted data [8] are presented in Fig. 4 and Table 4. It is seen that a reasonable agreement for the thermodynamic properties is obtained between the calculated results and the predicted data. Fig. 5 shows the calculated

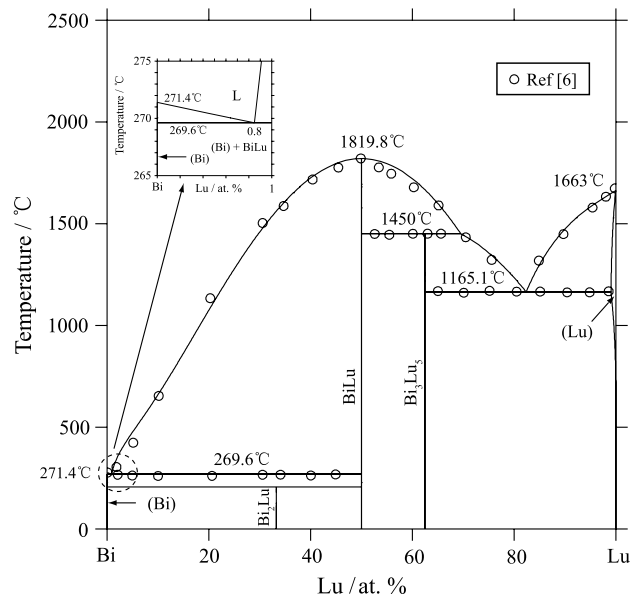

Fig. 3. Calculated phase diagram of the Bi–Lu system compared with the experimental data [6].

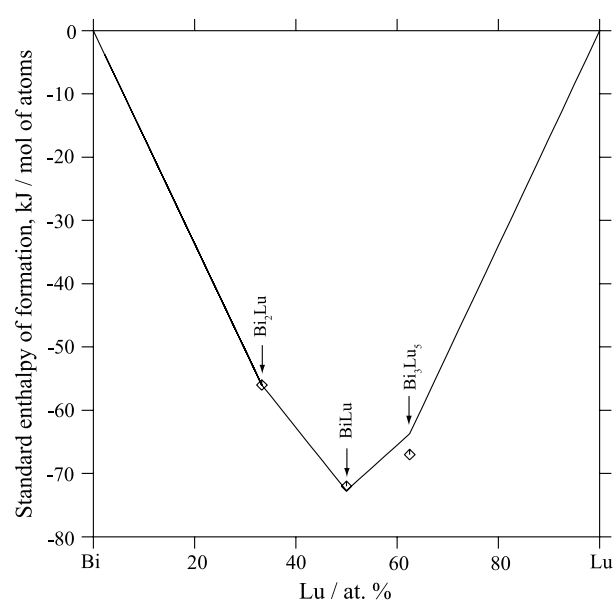

Fig. 4. Calculated standard enthalpies of formation of intermetallic compounds with reference states of Bi (rhombohedral) and Lu (hcp) phases in the Bi–Lu system compared with the predicted data [8].

Table 5
Experimental and calculated invariant reactions in the Lu–Sb system.

Invariant reaction	Reaction type	Composition (at.% Sb)			Temperature (°C)	Reference
L ↔ (Lu) + Lu_3Sb	Eutectic	~15.5	<2.0	25	1200	[10]
		13.2	1.9	25	1198.4	This work
L + Lu_5Sb_3 ↔ Lu_3Sb	Peritectic		37.5	25	1290	[10]
		16.6	37.5	25	1290	This work
L + αLuSb ↔ Lu_5Sb_3	Peritectic		50.0	37.5	1570	[10]
		24.6	50.0	37.5	1570.3	This work
αLuSb ↔ βLuSb	Polymorphic		50.0		1870	[10]
			50.0		1869.7	This work
L ↔ βLuSb	Congruent		50.0		2180	[10]
			50.0		2180.1	This work
L + αLuSb ↔ $LuSb_2$	Peritectic		50.0	66.7	930	[10]
		93.5	50.0	66.7	930.1	This work
L ↔ $LuSb_2$ + (Sb)	Eutectic	~99	66.7	100	625	[10]
		99.2	66.7	100	627.5	This work

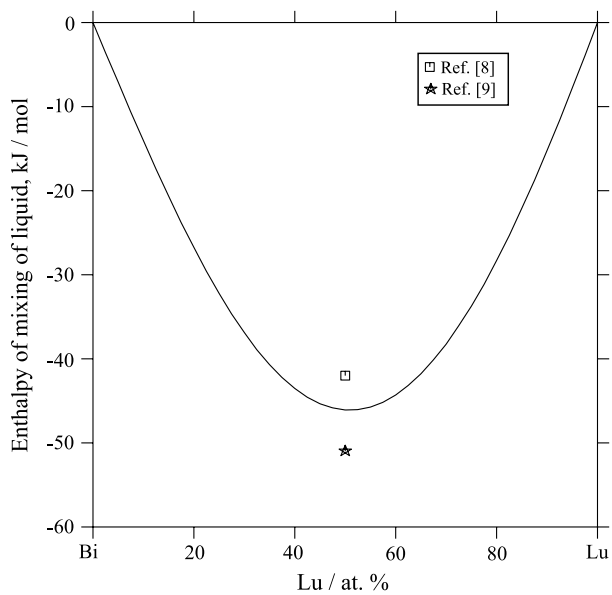

Fig. 5. Calculated enthalpies of mixing of the liquid phase at 2000 °C with reference states of pure Bi (liquid) and Lu (liquid) compared with the predicted data [8,9].

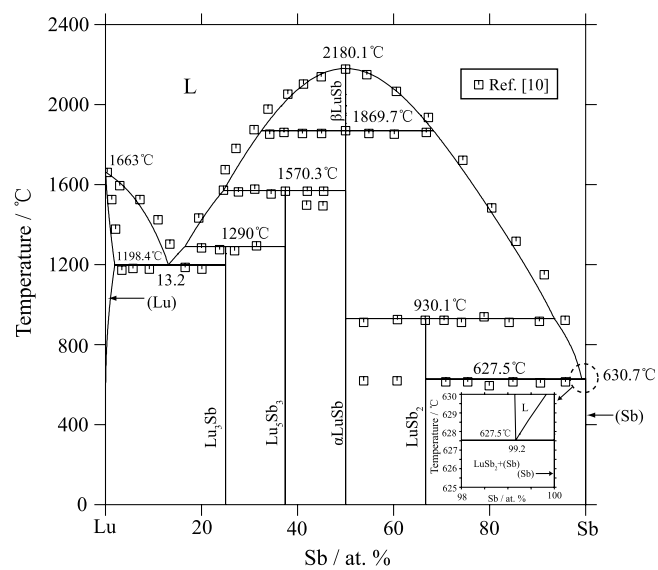

Fig. 6. Calculated phase diagram of the Lu–Sb system compared with the experimental data [10].

enthalpies of mixing of the liquid phase at 2000 °C with reference states of pure Bi (liquid) and Lu (liquid) in the Bi–Lu binary system. The calculated value at 50 at.% Lu is −46.2 kJ/mol, which is close to the predicted values of −42 kJ/mol [8] and −51 kJ/mol [9]. Therefore, the calculated enthalpies of mixing of the liquid phase in the Bi–Lu system are reasonable.

4.2. The Lu–Sb system

The calculated Lu–Sb phase diagram compared with the experimental data [10] is shown in Fig. 6, and the calculated compositions and temperatures for the invariant reactions compared with the selected experimental data are listed in Table 5. It is seen that the calculated results are in reasonable agreement with the experimental data. The calculated solid solubility of Sb in the hcp (Lu) solution phase is 1.9 at.% Sb, which is in agreement with the experimental data (<2.0 at.% Sb) [10] and considered reasonable. A set of complete self-consistent thermodynamic parameters describing the Gibbs free energy of each phase in the Lu–Sb system is given in Table 6.

The calculated standard enthalpies of formation of intermetallic compounds with reference states of the Lu (hcp) and Sb (rhombohedral) phases compared with the experimental data are presented in Fig. 7 and Table 7. It is seen that the calculated standard enthalpy

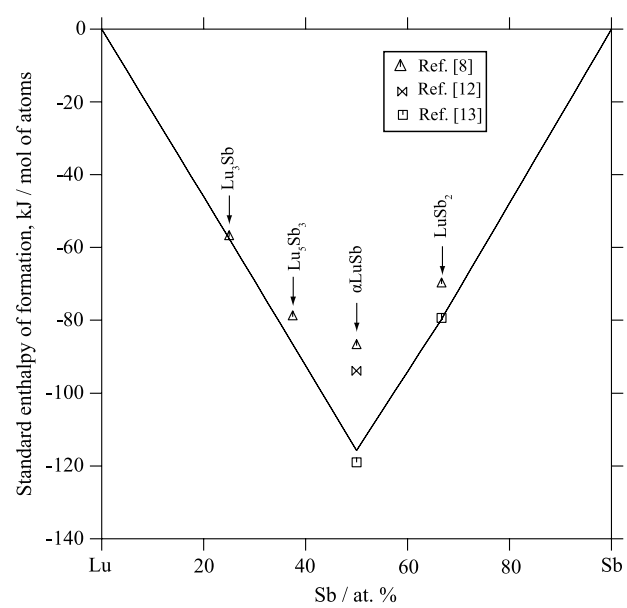

Fig. 7. Calculated standard enthalpies of formation of intermetallic compounds with reference states of Bi (rhombohedral) and Lu (hcp) phases in the Lu–Sb system compared with the experimental data [8,12,13].

Table 6
Thermodynamic parameters in the Lu–Sb system assessed in the present work.

Parameters in each phase (J/mol)
Liquid phase, format (Lu, Sb)
${}^{0}L^{Liq}_{Lu,Sb} = -229659 - 2.7T$
${}^{1}L^{Liq}_{Lu,Sb} = -12000 - 2.8T$
${}^{2}L^{Liq}_{Lu,Sb} = 1000 + 23T$
Hcp phase, format (Lu, Sb)
${}^{0}L^{Liq}_{Lu,Sb} = -174500$
Lu_3Sb phase, format $(Lu)_{0.75}(Sb)_{0.25}$
$\Delta^0 G_f^{Lu_3Sb} = -57500 + 2.355T$
Lu_5Sb_3 phase, format $(Lu)_{0.625}(Sb)_{0.375}$
$\Delta^0 G_f^{Lu_5Sb_3} = -86560 + 4.89T$
αLuSb phase, format $(Lu)_{0.5}(Sb)_{0.5}$
$\Delta^0 G_f^{\alpha LuSb} = -115746 + 9.414T$
βLuSb phase, format $(Lu)_{0.5}(Sb)_{0.5}$
$\Delta^0 G_f^{\beta LuSb} = -115500 + 9.299T$
$LuSb_2$ phase, format $(Lu)_{0.333}(Sb)_{0.667}$
$\Delta^0 G_f^{LuSb_2} = -79709 + 6.293T$

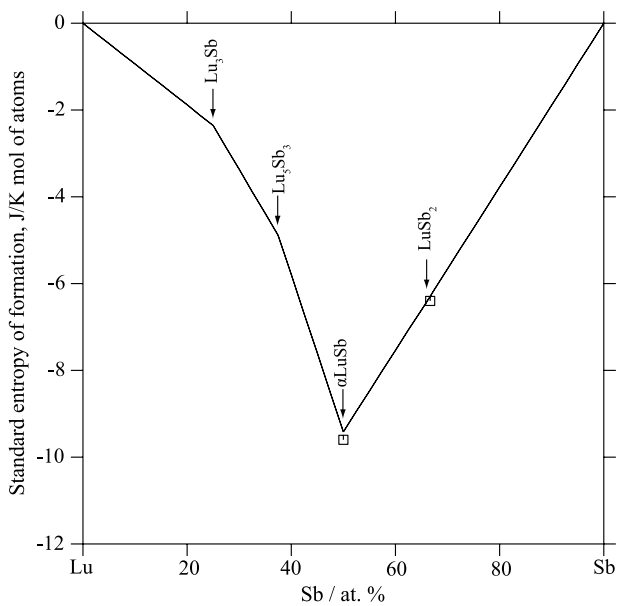

Fig. 8. Calculated standard entropies of formation of intermetallic compounds with reference states of Bi (rhombohedral) and Lu (hcp) phases in the Lu–Sb system compared with the experimental data [13].

of formation of the Lu_5Sb_3 phase is slightly more negative than the value predicted by Colinet et al. [8]. By considering the agreements of the most of the experimental data including the data of phase equilibria and thermodynamic properties, we think that this calculated result can be accepted. The calculated standard entropies of formation of intermetallic compounds with reference states of the Lu (hcp) and Sb (rhombohedral) phases compared with the experimental data are presented in Fig. 8 and Table 8. It is seen that a reasonable agreement of the thermodynamic properties is obtained between the calculated results and the experimental data.

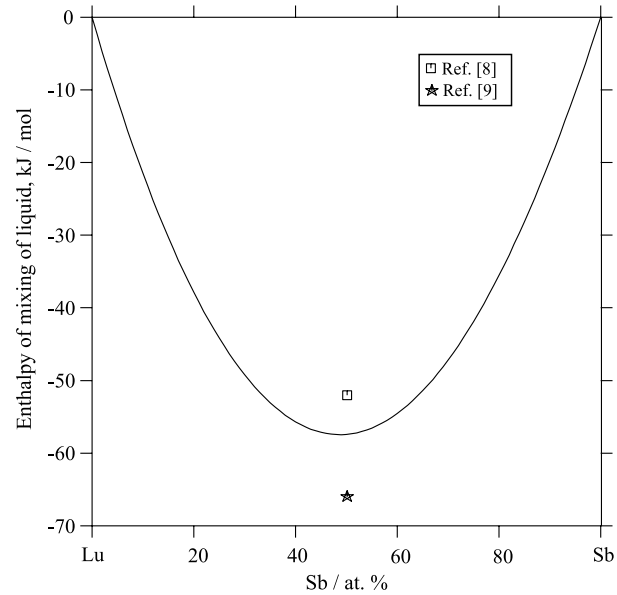

Fig. 9. Calculated enthalpies of mixing of liquid phase at 2200 °C with reference states of pure Lu (liquid) and Sb (liquid) compared with the predicted data [8,9].

Table 8
Comparison of experimental and calculated values for the standard entropies of formation (J/K mol of atoms) of intermetallic compounds in the Lu–Sb system.

Phase	Experimental [13]	Calculated
Lu_3Sb		−2.355
Lu_5Sb_3		−4.89
αLuSb	−9.6	−9.414
$LuSb_2$	−6.4	−6.293

The reference states of pure Lu and Sb are hcp and rhombohedral phases, respectively.

Fig. 9 shows the calculated enthalpies of mixing of liquid phase at 2200 °C with reference states of pure Lu (liquid) and Sb (liquid) in the Lu–Sb binary system. The calculated value at 50 at.% Sb is −57.415 kJ/mol, which is close to the values of −52 kJ/mol [8] and −66 kJ/mol [9]. Therefore, the calculated enthalpies of mixing of the liquid phase in the Lu–Sb system are reasonable.

5. Conclusions

The phase diagrams in the Bi–Lu and Lu–Sb binary systems were thermodynamically assessed based on experimental data including phase equilibria and thermodynamic data, and most of the experimental information can be satisfactorily reproduced on the basis of the optimized thermodynamic parameters.

Acknowledgments

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Table 7
Comparison of experimental and calculated values for the standard enthalpies of formation (kJ/mol of atoms) of intermetallic compounds in the Lu–Sb system.

Phase	Predicted [8]	Experimental [12]	Experimental [13]	Calculated
Lu_3Sb	−57			−57.5
Lu_5Sb_3	−79			−86.56
αLuSb	−87	−94	−119	−115.746
$LuSb_2$	−70		−79.3	−79.7

The reference states of pure Lu and Sb are hcp and rhombohedral phases, respectively.

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Appendix. Supplementary data

Supplementary material related to this article can be found online at doi:10.1016/j.calphad.2011.05.003.

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